Answer key: Lesson 6, Tens Jump, Ones Hop

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Mei has these sticks. She gets 1 more bundle of ten.

Answer: 3 bundles, 6 singles, 36 sticks

The picture shows 2 bundles and 6 singles, which is 26. One more bundle makes 3 bundles, and the 6 singles stay. 30 and 6 is 36.

Watch for: Adding the bundle to the singles (27).; Forgetting the old bundles (16).

S2. Draw 3 bundles of ten and 4 single sticks in the box. Then draw 2 more bundles of ten. Write the number.

Answer: 54 (5 bundles drawn and 4 singles)

3 bundles and 4 singles is 34. Two more bundles add 20, so there are 5 bundles and 4 singles: 54.

Watch for: Adding the 2 to the singles (36).; Drawing 2 sticks instead of 2 bundles of ten.

S3. Jump forward. Write the number.

Answer: 33, 63, 81, 67

(a) 2 tens and 1 ten make 3 tens: 33. (b) 2 tens and 4 tens make 6 tens: 63. (c) 5 tens and 3 tens make 8 tens: 81. (d) 7 is 7 ones; add 6 tens to get 67.

Watch for: Changing the ones digit (23 + 40 = 27).; (d) writing 13 (7 + 6).

S4. Jump back, or hop. Write the number.

Answer: 24, 38, 55, 69

(a) 3 tens take away 1 ten is 2 tens: 24. (b) 5 tens take away 2 tens is 3 tens: 38. (c) 3 is a hop: 8 ones take away 3 is 5 ones: 55. (d) 7 is a hop forward: 2 and 7 make 9 ones: 69.

Watch for: Taking the 3 from the tens digit (58 - 3 = 28).; Hopping the wrong way in (d).

S5. Ali has 68 pencils. He gives away 3 boxes of ten pencils. How many pencils are left?

Answer: 38 pencils

3 boxes of ten is 30 pencils. 68 - 30 = 38: jump back 3 tens (58, 48, 38). The 8 ones stay.

Watch for: Taking away 3 (65).; Adding instead of taking away (98).

S6. Siti says 45 + 20 = 47. Is Siti right? Write the right answer. Say what she mixed up.

Answer: No. 65. She hopped 2 ones instead of jumping 2 tens.

20 is 2 tens, so Siti should jump two tens: 45, 55, 65. She added 2 ones and got 47. A jump changes the tens digit.

Watch for: Saying she is right because 20 has a 2 in it.; Not saying what she mixed up (any clear sentence about tens and ones is fine).

S7. Start at 36. Draw 2 jumps of ten on a number line in the box. Write where you land.

Answer: 56 (two jumps of ten from 36: 46, then 56)

36 + 10 = 46, then 46 + 10 = 56. Two jumps of ten land on 56. The 6 ones never change.

Watch for: Drawing 2 small hops (38).; Changing the ones digit.

Think harder (page 3)

T1. Work out 34 + 25. Jump the tens first, then hop the ones. Show your jumps and hops.

Answer: 59 (jump 20 to 54, then hop 5 to 59)

25 is 2 tens and 5 ones. 34 + 20 = 54. Then 54 + 5 = 59. Check: 3 tens and 2 tens are 5 tens; 4 ones and 5 ones are 9 ones.

Watch for: Adding only the tens and stopping at 54.; Adding 2 + 5 as 7 tens (74).

T2. Work out 68 - 24. Jump back the tens first, then hop back the ones.

Answer: 44 (jump back 20 to 48, then hop back 4 to 44)

24 is 2 tens and 4 ones. 68 - 20 = 48. Then 48 - 4 = 44.

Watch for: Taking 24 away from the ones only.; Hopping forward instead of back.

T3. Frog A starts on 12. It jumps 3 tens, then hops 4. Frog B starts on 12. It hops 4 first, then jumps 3 tens. Do they land on the same number? How do you know?

Answer: Yes, both land on 46 (30 and 4 are added in either order)

Frog A: 12 + 30 = 42, then 42 + 4 = 46. Frog B: 12 + 4 = 16, then 16 + 30 = 46. They land on the same number: adding the same jumps and hops in a different order gives the same total.

Watch for: Saying Frog B lands on 16 because the hops came first.; Adding 3 + 4 and landing on 19.

T4. Ali starts on a number. He jumps forward 3 tens and lands on 78. Where did he start?

Answer: 48 (jump back 3 tens from 78)

Undo the 3 jumps of ten: 78, 68, 58, 48. Check: 48 + 30 = 78.

Watch for: Adding 30 to 78 (108).; Taking away only 3 (75).

Go further (page 4)

G1. Mei is on 7. She makes jumps of ten until she gets to 97. How many jumps does she make? Write each number she lands on.

Answer: 9 jumps: 17, 27, 37, 47, 57, 67, 77, 87, 97

7 + 10 = 17, then 27, 37, 47, 57, 67, 77, 87, 97. That is 9 jumps. The ones digit stays 7. Check: 97 is 9 tens more than 7.

Watch for: Saying 10 jumps (counting the start).; Stopping at 87.

G2. Mei goes from 7 to 97 again. This time she may use jumps of 20 as well as jumps of 10. What is the fewest jumps she can make? Write her jumps.

Answer: 5 jumps: four jumps of 20 and one jump of 10 (any order)

She must travel 9 tens. Four jumps of 20 make 8 tens. One jump of 10 makes the last ten. That is 5 jumps. Four jumps can travel at most 80, which only reaches 87, so 5 is the fewest.

Watch for: Saying 4 (80 only reaches 87).; Saying 9 (only using jumps of 10).

G3. Start at 25. Jump forward 3 tens. Jump back 1 ten. Jump forward 2 tens. Then hop forward 4. Where do you land? Write each landing number.

Answer: 69 (25, 55, 45, 65, 69)

25 + 30 = 55. 55 - 10 = 45. 45 + 20 = 65. 65 + 4 = 69. Short way: 3 - 1 + 2 = 4 tens forward, so 25 + 40 = 65, then 4 more.

Watch for: Jumping forward on the jump back.; Forgetting the hop of 4 (65).

Investigation (page 5): Jump road

From 6 you land on 16, 26, 36, 46, 56 and so on: every number ends in 6, so you never land on 50. Jumps of ten never change the ones digit. Only a start number that ends in 0 (0, 10, 20, 30 or 40) can land on 50.
Look for: A child who tests more than one start number, says the ones digit stays the same, and gives a reason using tens and ones (a jump is whole bars). A child who says start 20 can land on 50 because 20 + 30 = 50.

Exit ticket (page 5)

E1. Jump forward. Write the number.
Answer: 76
4 tens and 3 tens make 7 tens. The 6 ones stay: 76.
Watch for: Writing 49 (adding 3 to the ones).; Writing 16.

E2. Jump back, or hop. Write the numbers.
Answer: 44 and 71
(a) 30 is 3 tens: jump back to 44. (b) 3 is a hop: hop back to 71.
Watch for: Doing the same for both parts.; (b) writing 44.

E3. Jump the tens, then hop the ones. Write the number.
Answer: 59
28 is 2 tens and 8 ones. 31 + 20 = 51, then 51 + 8 = 59.
Watch for: Stopping at 51.; Writing 39.

Sneak-peek puzzles (not in the workbook)

1. 36 stars.
2. No. 43 + 20 = 63. 20 is 2 tens, so the tens digit changes, not the ones.